

Project Design Phase

Problem – Solution Fit Template

Date	07 July 2026
Team ID	SWTID-2026-2247
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.
- Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- **Understand the existing situation in order to improve it for your target group.**

Template:

Problem – Solution Fit Canvas
Emotion Detection and Learning Support Engine

Customer Segment(s) School/college students (age 13-22) in online & hybrid classrooms; secondarily teachers and parents who support them		
Problem (Top 3) 1. Confusion/boredom/stress goes unnoticed until grades drop 2. Teachers can't track every student's emotion in real time 3. Adaptive platforms react only to scores, not feelings	Root Cause Emotional signals (expression, tone, hesitation) are invisible in digital learning; no feedback loop links emotion to content pacing/difficulty	Solution (Top 3 Features) 1. Real-time multimodal emotion detection inside existing LMS/ video class 2. Adaptive engine that adjusts pace/content on frustration 3. Teacher/parent dashboard with real-time alerts
Existing Alternatives Periodic mood surveys, manual teacher observation, quiz-score-only adaptive platforms, post-hoc counselling after issues surface	Customer Constraints Limited bandwidth/webcam access, privacy concerns around recording minors, teachers' limited time to watch dashboards	Trigger to Act Engagement/attention visibly drops mid-lesson; a parent-teacher meeting reveals a hidden struggle; repeated quiz failures or distress chat messages
Behaviour Today Students mute video and disengage silently; teachers ask "any questions?" with no real signal; parents rely on report cards / PTA meetings	Channels of Behaviour School LMS (Google Classroom, Moodle), video tools (Zoom, Teams), student mobile apps, WhatsApp/SMS updates to parents	Emotions (Before / After) Before: frustration, isolation, anxiety about falling behind After: reassurance, relief, renewed motivation, feeling supported without having to ask

Problem – Solution Fit Details:

Customer Segment(s): School/college students (age 13-22) in online & hybrid classrooms; secondarily teachers and parents who support them		
Problem (Top 3) ● Confusion, boredom, or stress goes unnoticed until grades drop ● Teachers cannot track every student's emotional state in real time ● Adaptive platforms react only to quiz scores, not to how the student feels	Root Cause / Constraints ● Emotional signals (expression, tone, hesitation) are invisible in digital learning ● No feedback loop links emotion to content pacing or difficulty ● Limited bandwidth/webcam access and privacy concerns around recording minors	Solution (Top 3 Features) ● Real-time multimodal emotion detection built into the existing LMS/video class ● Adaptive learning engine that adjusts pace/content the moment frustration is detected ● Teacher/parent dashboard with real-time alerts and weekly trend reports
Existing Alternatives ● Periodic self-report mood surveys / check-ins ● Manual teacher observation in class ● Quiz-score-only adaptive learning platforms ● Post-hoc counselling after a student is already flagged	Channels of Behaviour ● School LMS: Google Classroom, Moodle ● Video conferencing: Zoom, MS Teams ● Student mobile apps used outside class ● WhatsApp/SMS updates to parents	Trigger to Act ● Engagement/attention visibly drops mid-lesson ● A parent-teacher meeting reveals a hidden struggle ● Repeated quiz failures or a distress message in chat
Behaviour Today ● Students mute video and disengage silently ● Teachers ask "any questions?" and get no real signal ● Parents rely on report cards / periodic PTA meetings		Emotions (Before / After) ● Before: frustration, isolation, anxiety about falling behind ● After: reassurance, relief, renewed motivation, feeling supported without having to ask